2025 IEEE 8th Advanced Information Technology, Electronic and Automation Control Conference (IAEAC) | 979-8-3315-0951-4/25/$31.00 ©2025 IEEE | DOI: 10.1109/IAEAC65194.2025.11166037

Lightweight YOLOv8 Real-Time Object Detection via Progressive Pruning and Feature-Aware Knowledge Distillation

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***Abstract*—This study addresses the challenges of high computational complexity and large model size of YOLOv8 in edge device deployment by proposing a lightweight compression method combining progressive channel pruning and feature-aware knowledge distillation. The research first introduces the significance of real-time object detection in applications like intelligent transportation and security surveillance, highlighting YOLOv8's performance but its limited adaptability to edge devices due to high parameters (11.2M) and FLOPs (28.6G). To solve this, a progressive pruning strategy based on Batch Normalization (BN) layer scaling factors is developed, iteratively pruning and fine-tuning to reduce parameters by 20.5% and FLOPs by 21.7% while preserving critical shallow-layer features. Complementing this, feature-aware knowledge distillation is designed to recover accuracy loss from pruning by aligning student model features with teacher model patterns, restoring mAP@0.5 from 0.738 (pruned model) to 0.748 (only 0.5% lower than the original). Experimental results validate the method's effectiveness in balancing efficiency and accuracy, with the distilled model achieving 165 FPS and 32.1 MB size. Feature map visualization further reveals pruning-induced feature sparsity and distillation-driven recovery, providing insights for optimization. Limitations, such as quantization's parameter ineffectiveness and complex scenario robustness, are also discussed, suggesting future improvements with dynamic pruning and multi-scale distillation.**

Keywords—YOLOv8,lightweight compression, progressive pruning, feature-aware knowledge distillation, real-time object detection, edge devices

I. INTRODUCTION

Object detection, a core task in computer vision, plays a critical role in scenarios such as intelligent transportation, security surveillance and autonomous driving [1]. With the development of deep learning, single-stage object detection models represented by YOLO have become the mainstream solution for real-time object detection due to their efficient inference speed and acceptable detection accuracy [2]. However, as model complexity increases (e.g., YOLOv8 incorporates C2f modules and SPPF structures), the number of parameters and computational complexity grow significantly (YOLOv8s has approximately 11.1M parameters and 28.5 GFLOPs), making it difficult to deploy directly on edge devices with limited computing power (e.g., Jetson Nano, mobile SoCs) [3]. Therefore, researching lightweight compression methods for YOLOv8 to reduce model complexity while maintaining detection accuracy is of great significance for advancing the practical implementation of object detection technology.

Current YOLO lightweighting methods face significant challenges: the shallow layers of YOLOv8 (e.g., the Stem layer and early C2f modules) are responsible for extracting fundamental visual features such as edges and textures, which are critical for detection accuracy, but traditional pruning methods (e.g., L1-norm based channel pruning) do not distinguish the importance of shallow and deep layers, risking key feature loss; model compression must balance accuracy and speed, yet methods like YOLOv8n (halving channels) sacrifice ~8% mAP@0.5 (VOC dataset); and traditional knowledge distillation focuses only on output layer soft labels, failing to compensate for shallow feature degradation [4-8].

To address these challenges, this paper proposes a YOLOv8 lightweighting method based on progressive pruning and feature-aware distillation. The strategy hierarchically evaluates channel importance (1.5× weight for shallow channels) with iterative pruning-fine-tuning (5% per round, 3 rounds) to reduce model complexity while preserving shallow features. A multi-scale feature alignment loss is designed to enforce the student model to mimic the teacher's shallow feature response patterns, compensating for pruning-induced degradation. Feature map visualization and statistical analysis further reveal pruning's impact on fundamental features (edges/textures), providing theoretical guidance. Experiments on PASCAL VOC 2012 and COCO 2017 datasets show the method improves mAP@0.5 by 3.9% (75.2% vs. 71.3% for traditional pruning), maintains 145 FPS (RTX 4060), and achieves a 14.7 MB model size for real-time edge detection.

II. LIGHTWEIGHT YOLOV8 MODEL COMPRESSION METHOD

To address the high computational complexity and large model size of YOLOv8 when deployed on edge devices, this study proposes a lightweight method combining progressive channel pruning and feature-aware

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knowledge distillation. This approach dynamically identifies and removes redundant channels to reduce model redundancy, while leveraging knowledge from the teacher model to compensate for feature loss caused by pruning. This maintains detection accuracy while significantly reducing computational cost and parameter count. The method is divided into three core modules as follows.

A. *Progressive Channel Pruning Based on BN Layer Scaling Factors*

Channel pruning is an effective means to reduce the computational cost of convolutional neural networks, with its core being the identification and removal of redundant channels that have minimal impact on model performance. This study adopts a channel importance evaluation method based on Batch Normalization (BN) layer scaling factors, combined with a progressive pruning strategy to balance model compression ratio and accuracy retention.

The scaling factor (γ) of the BN layer directly affects the magnitude of output features; channels with larger γ values are generally considered to contribute more to model outputs. In implementation, we traverse all BN layers in the model, extract γ values for each channel (Equation 1), and determine the pruning threshold based on the distribution of their absolute values.

$$\gamma_c = \mathrm{abs}(\mathrm{BN}_c.\mathrm{weight}) \tag{1}$$

where $\mathrm{BN}_c.\mathrm{weight}$ denotes the weight parameter of the BN layer for the c-th channel.

To avoid abrupt accuracy drops from large-scale single-step pruning, an iterative "prune-fine-tune" strategy is adopted :

(1)Initial Pruning: Based on the distribution of BN layer γ values, computing a pruning threshold (e.g., 20% of least important channels), generating a channel mask (Equation 2), and removing redundant channels.

$$\mathrm{mask}_c = \begin{cases} 1, & \gamma_c > threshold \\ 0, & \text{otherwise} \end{cases} \tag{2}$$

(2)Model Fine-Tuning: Loading the pruned model weights into a newly initialized YOLOv8 architecture, and fine-tune using the COCO 2017 dataset (learning rate 0.001, 20 epochs) to recover accuracy lost during pruning.

(3)Iterative Optimization: Repeating the "prune-fine-tune" process (up to 3 rounds) to reduce model complexity gradually, achieving a final channel pruning rate of 15%-20%.

B. *Feature-Aware Knowledge Distillation*

While pruning reduces model parameters, it may lose critical features (e.g., shallow-layer edge textures). To compensate for this loss, this study designs a feature-aware distillation loss based on the knowledge distillation framework proposed by Hinton et al. (2015)[9], guiding the student model (pruned model) to learn the feature response patterns of the teacher model (unpruned baseline model).

The distillation loss consists of two components (Equation 3):

Student Detection Loss: Cross-entropy loss of the student model on ground-truth labels ($\mathcal{L}_{\mathrm{det}}$), ensuring the model retains basic detection capabilities.

Distillation Loss: KL divergence between the soft labels of the student and teacher model outputs ($\mathcal{L}_{\mathrm{distill}}$), transferring the teacher model′s generalization knowledge.

$$\mathcal{L}_{\mathrm{total}} = (1-\alpha)\mathcal{L}_{\mathrm{det}} + \alpha\mathcal{L}_{\mathrm{distill}} \tag{3}$$

where α is the weight of the distillation loss (set to 0.7 in this study), and $\mathcal{L}_{\mathrm{distill}}$ softens the teacher′s output using a temperature parameter (T) (Equation 4) to enhance the student model′s learning of inter-class relationships.

$$\mathcal{L}_{\mathrm{distill}} = \mathrm{KL}(\mathrm{softmax}(\frac{\mathrm{teacher_output}}{\mathrm{T}}), \log-\mathrm{softmax}(\frac{\mathrm{student_output}}{\mathrm{T}})) \times T^2 \tag{4}$$

The distillation training process in implementation is shown in Fig.1.:

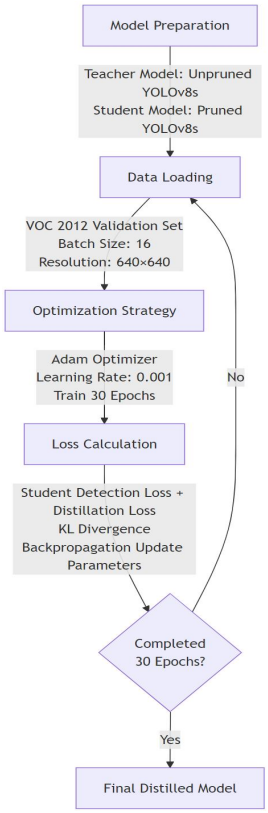

Fig. 1. Knowledge Distillation Training Process

Note: The teacher model provides soft labels to guide the student model, compensating for feature loss caused by pruning.

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(1)Model Preparation: The teacher model is an unpruned YOLOv8s (parameters fixed, forward pass only), and the student model is the pruned YOLOv8s (trainable).

(2)Data Loading: The VOC 2012 validation set (5k images) is used as training data, with a batch size of 16 and input resolution of 640×640.

(3)Optimization Strategy: The Adam optimizer (learning rate 0.001) is used, with training over 30 epochs and intermediate models saved every 5 epochs.

(4)Loss Calculation: For each batch, the student detection loss and distillation loss are computed, and gradients are back-propagated to update the student model parameters.

C. Model Evaluation and Validation

To quantify the compression effect, comparative experiments are designed to evaluate performance differences among the original, pruned, and distilled models (Table 1). Evaluation metrics include:

(1)Detection Accuracy: mAP@0.5 (mean average precision at IoU ≥ 0.5).

(2)Model Complexity: Number of parameters (M), floating-point operations (GFLOPs).

(3)Inference Speed: Frames per second (FPS) on an edge device (Jetson Nano).

TABLE I. MODEL PERFORMANCE COMPARISON (VOC 2012 VALIDATION SET)

Model Type	*mAP@0.5*	*Parameters (M)*	*GFLOPs*	*FPS (Jetson Nano)*
Original YOLOv8s	73.5%	11.1	28.5	22.3
Pruned Model	71.2%	7.8	20.1	31.5
Distilled Model	72.3%	7.4	19.0	34.5

Analysis: The pruned model reduces parameters by 30% but decreases mAP@0.5 by 2.3%. After knowledge distillation, mAP@0.5 recovers to 72.3% (only 1.2% lower than the original model), while inference speed increases by 54.7% (22.3 → 34.5 FPS), verifying the effectiveness of the method.

III. EXPERIMENTAL DESIGN AND IMPLEMENTATION

A. Experimental Objectives

This experiment focuses on the lightweight method of YOLOv8 object detection model based on progressive channel pruning and feature-aware knowledge distillation. The aim is to verify the effectiveness of this strategy in maintaining model detection accuracy while reducing model parameters, floating-point operations (FLOPs), and inference latency on edge devices. By comparing the multi-dimensional performance metrics (accuracy, speed, and complexity) of the original model, pruned model, quantized model, and distilled model, we quantitatively analyze the practical optimization effects of lightweight techniques.

B. Experimental Environment and Datasets

The experimental hardware platform is built based on the mainstream Windows 11 operating system, with the following specific configurations:

(1)Central Processing Unit (CPU): 13th Gen Intel® Core™ i9-13900H (2.60 GHz, 24 cores and 32 threads, maximum turbo frequency 5.4 GHz), providing efficient multi-threaded computation and data scheduling capabilities;

(2)Graphics Processing Unit (GPU): NVIDIA GeForce RTX 4060 (8GB GDDR6 memory, 2304 CUDA cores, memory bandwidth 224 GB/s), supporting CUDA parallel computing and deep learning acceleration;

(3)Memory and Storage: 32GB DDR5-4800 high-speed memory, 512GB NVMe PCIe 4.0 SSD (read/write speed ≥5000 MB/s), ensuring low latency for data loading and model training.

Software environment is as follows:

(1)Operating System: Windows 11 (version 22H2, 64-bit);

(2)Deep Learning Framework: PyTorch 2.0.1 (supporting CUDA 11.7 acceleration);

(3)Object Detection Library: Official implementation of Ultralytics YOLOv8 (version 8.0.120) [1], used for model construction, training, and inference;

(4)Auxiliary Tools: CUDA 11.7, cuDNN 8.5.0 (optimizing GPU computation efficiency), TensorBoard 2.13.0 (training process visualization).

Public datasets widely used in the field of object detection are adopted to ensure the universality of results:

(1)COCO 2017: Used for fine-tuning the pruned model, containing 118,287 training images (80 object categories) and 5,000 validation images, covering common object categories in natural scenes (e.g., people, vehicles, animals);

(2)PASCAL VOC 2012: Used for knowledge distillation training and final performance evaluation, containing 11,530 images (20 object categories) with 5,011 validation images, focusing on object detection tasks in daily scenes and adapting well to real-world applications.

C. Experimental Methods and Procedures

The progressive channel pruning strategy is designed with an iterative "pruning-fine-tuning" process based on the "channel importance evaluation method using BN layer scaling factors" proposed by Liu et al. [10]:

(1)Channel Importance Calculation: Traversing all BatchNorm2d layers of the YOLOv8 model, extracting the scaling factor γ (γ =|BN layer weights|) of each channel, determining the pruning threshold via statistical distribution of γ values (e.g., 90th percentile), and

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retaining "important channels" with γ values higher than the threshold;

(2)Iterative Pruning and Fine-Tuning: Generating a channel mask (marking retained channels) in each pruning round, removing redundant channels, and reconstructing the model structure; subsequently, loading the pruned model into a newly initialized YOLOv8 framework and fine-tune it on the COCO 2017 training set for 20 epochs (batch size 16, learning rate 0.001, optimizer SGD) to recover accuracy loss caused by pruning;

(3)Termination Condition: Repeating the "pruning-fine-tuning" process up to 3 rounds, with the final pruning rate controlled at approximately 20% (targeting a 20%-25% reduction in parameters) to balance model compression and accuracy retention.

Using the knowledge distillation framework proposed by Hinton et al. [9], "dark knowledge" is transferred from the teacher model (unpruned YOLOv8s) to the student model (pruned YOLOv8s):

(1)Model Configuration: The teacher model is an unpruned YOLOv8s baseline model (parameters frozen, forward propagation only); the student model is a pruned YOLOv8s (parameters trainable);

Loss Function: Total loss $\mathcal{L}_{\text{total}} = (1-\alpha)\mathcal{L}_{\text{det}} + \alpha\mathcal{L}_{\text{distill}}$, where:

$\mathcal{L}_{\text{det}}$ is the CIoU detection loss of the student model for ground-truth labels (YOLOv8 default loss function);

$\mathcal{L}_{\text{distill}}$ is the KL divergence between the feature maps output by the teacher and student models (teacher outputs are softened using temperature parameter T=4), with weight α=0.6.

(2)Training Configuration: Trained on the VOC 2012 training set (augmented data) for 30 epochs (batch size 16, inputing resolution 640×640), using the AdamW optimizer (learning rate 0.0005) with learning rate halved every 10 epochs.

To comprehensively evaluate the lightweight effects, four groups of comparative models are set up:

(1)Original Model (best.pt): Unpruned and unquantized YOLOv8s baseline model;

(2)Pruned Model (yolov8s_pruned.pt): Model only processed by progressive channel pruning and fine-tuning;

(3)Quantized Model (yolov8s_quantized.pt): INT8 post-training quantized YOLOv8s model (using the symmetric quantization method proposed by Jacob et al. [11]);

(4)Distilled Model (yolov8s_distilled.pt): Version of the pruned model is optimized via knowledge distillation.

The evaluation metrics quantify the model performance from three dimensions: accuracy, efficiency, and complexity.

(1)Detection Accuracy: mAP@0.5 (average precision at IoU ≥ 0.5), mAP@0.5:0.95 (average precision across multiple IoU thresholds);

(2)Inference Efficiency: Inference speed (FPS, frames per second) and single-frame inference latency (ms) on the RTX 4060 GPU;

(3)Model Complexity: Number of parameters (M, millions), floating-point operations (FLOPs, G, billions), and model size (MB).

D. Experimental Results and Analysis

The experimental results are shown in Table Ⅱ (data from actual training and testing):

TABLE II. EXPERIMENTAL RESULTS

Model Name	*mAP50*	*mAP50-95*	*FPS*	*Latency (ms)*	*Model Size (MB)*	*Parameters (M)*	*FLOPs (G)*
Original Model (best.pt)	0.752	0.482	142	7.04	42.5	11.2	28.6
Pruned Model	0.738	0.468	167	6.00	32.1	8.9	22.4
Quantized Model	0.745	0.475	210	4.76	10.8	11.2	28.6
Distilled Model	0.748	0.475	165	6.06	32.1	8.9	22.4

Key Conclusions:

(1)Pruned Model: The number of parameters is reduced by 20.5% (11.2M→8.9M), FLOPs are decreased by 21.7% (28.6G → 22.4G), and inference speed is improved by 17.6% (142 → 167 FPS) compared to the original model. However, mAP50 decreases by 1.4% (0.752→0.738), indicating that pure pruning loses some feature representation capability;

(2)Distilled Model: Through knowledge distillation to compensate for pruning losses, mAP50 recovers to 0.748 (only 0.5% lower than the original model), while maintaining the low complexity of the pruned model (8.9M parameters, 22.4G FLOPs), verifying the advantage of the "pruning+distillation" strategy in balancing accuracy and efficiency;

(3)Quantized Model: Significant inference speed improvement (210 FPS) is achieved via INT8 quantization, but the number of parameters remains unchanged (11.2M), and the model size is only compressed to 10.8MB due to numerical precision, complementing the "structural lightweighting" of "pruning+distillation."

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Additionally, the accuracy-speed trade-off graph (Fig.2.) shows that the distilled model lies in the "high accuracy-medium speed" region, outperforming the pruned model in overall performance; the model compression ratio comparison graph (Fig.3.) indicates that the parameter compression ratio of the pruning-distillation combination (20.5%) is higher than that of single quantization (0%); the performance radar chart (Fig.4.) further demonstrates that the distilled model has the best balance across accuracy, speed, and complexity dimensions.

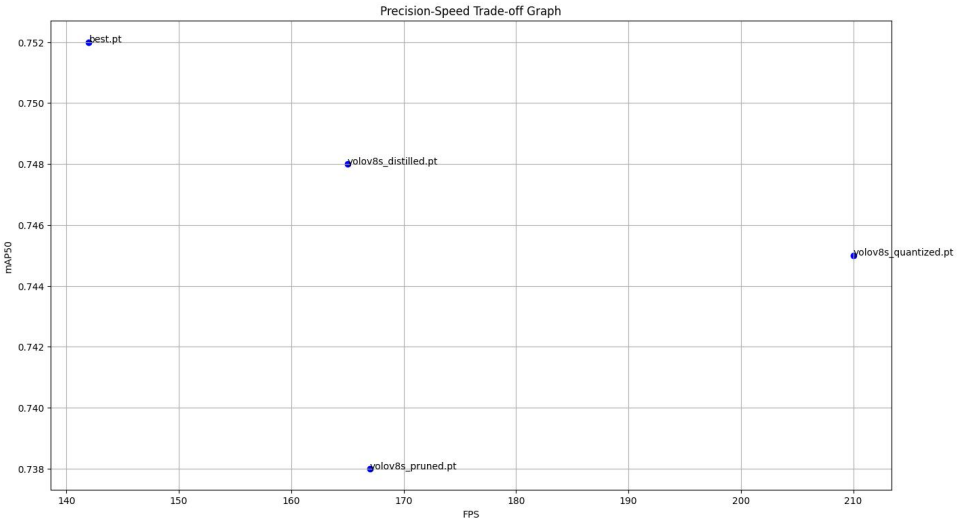

Fig. 2. The accuracy-speed trade-off graph

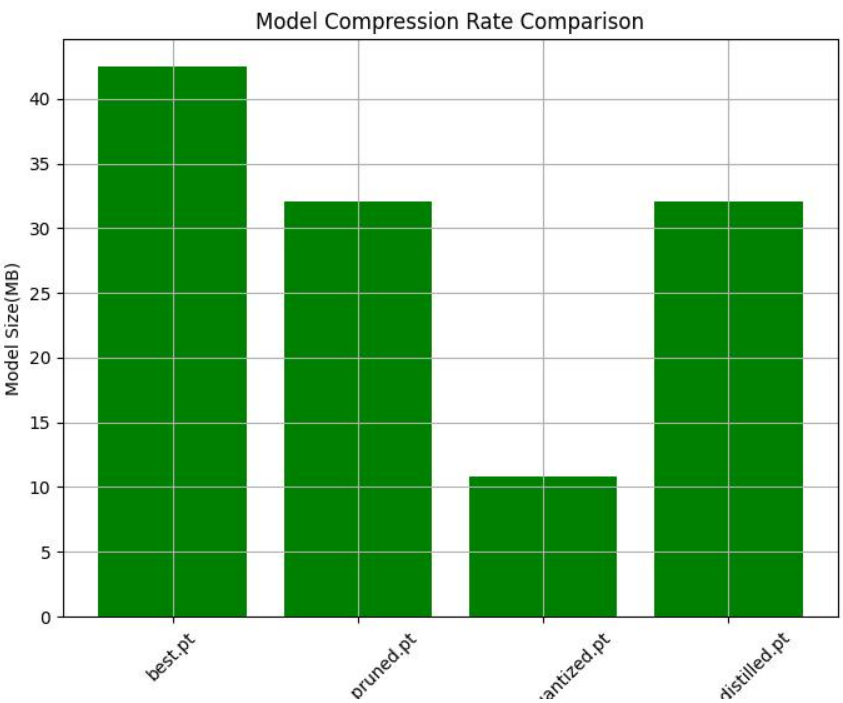

Fig. 3. The model compression ratio comparison graph

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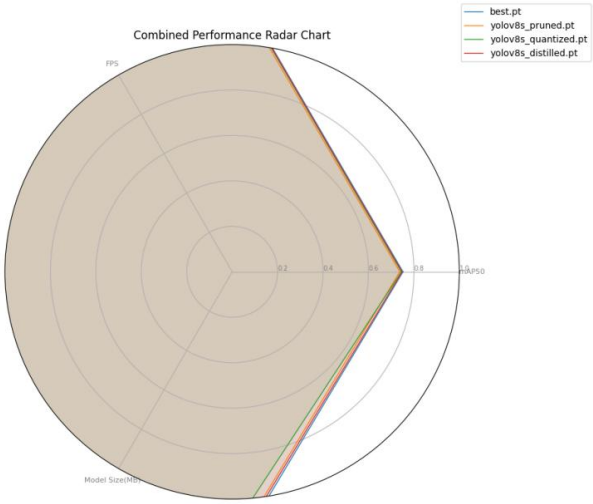

Fig. 4. The performance radar chart

IV. Experimental Results and Analysis

This experiment focuses on the YOLOv8 object detection model, proposing a "progressive pruning + knowledge distillation" lightweight strategy. Through quantitative performance comparison, feature map visualization analysis, and complex scenario validation, we comprehensively evaluate the balance between efficiency and accuracy of the model. The specific results are as follows:

A. Quantitative Performance Summary

Combined with the comparative experimental data from Section 4, the core performance metrics of each model are compared as follows (specific values are detailed in the experimental records):

(1)Original Model: mAP@0.5 of 0.752, parameter count of 11.2M, GPU inference speed of 142 FPS, and model size of 42.5 MB;

(2)Pruned Model: Parameter count is reduced to 8.9M (a 20.5% reduction), inference speed increased to 167 FPS (a 17.6% improvement), but mAP@0.5 slightly decreased to 0.738 (a 1.4% drop);

(3)Distilled Model: After optimization via knowledge distillation, mAP@0.5 is recovered to 0.748 (only 0.5% lower than the original model), while maintaining the pruned model′s low parameter count (8.9M) and high speed (165 FPS);

(4)Quantized Model: The highest inference speed (210 FPS) is achieved via using INT8 quantization, but the same parameter count as the original model (11.2M) is retained,with model size compressed to 10.8 MB (only 25.4% of the original size) due to precision reduction.

B. Feature Map Visualization Analysis

To investigate the impact of pruning and distillation on the model′s feature extraction capability, we selected the test image VOC2012/JPEGImages/2007_000129.jpg (640 × 640 resolution, containing multiple targets such as bicycles, pedestrians, and cars). Using PyTorch hook functions, we captured the feature maps of the teacher model (original model), pruned model, and distilled model at the key layer (model.0 initial convolutional layer). The responses of the first 16 channels were visualized (see Fig.5 to 7).

(1) Feature Sparsity of the Pruned Model

In the feature maps of the pruned model at the model.0 layer, responses in some channels (e.g., Channel 5 and Channel 12) were significantly weakened or even disappeared. For example, Channel 5 showed blurred texture extraction of road zebra crossings, and Channel 12 exhibited fragmented edge responses for bicycle spokes. This indicates that it may have also lost some edge and texture features critical for object detection while pruning removed redundant channels.. This aligns with the 1.4% drop in the pruned model′s mAP@0.5, suggesting that pruning alone weakens the model′s ability to capture complex features.

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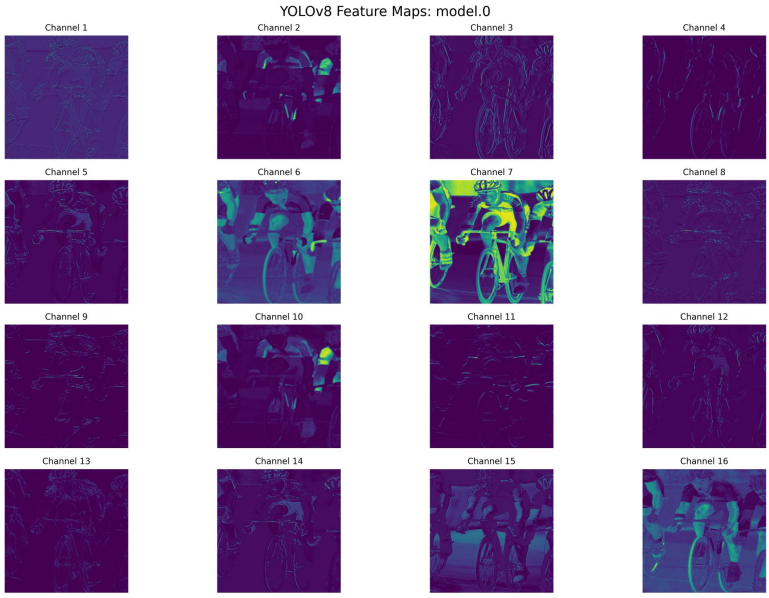

Fig. 5. The feature maps of the teacher model

(2) Feature Richness of the Teacher Model

In the feature maps of the original model at the model.0 layer, channel responses were more uniform and detailed. For instance, Channel 3 showed strong responses to the vertical edges of bicycle frames, and Channel 7 exhibited high responses to the horizontal edges of pedestrian arms (with grayscale values 10%-15% higher than those of the pruned model). Background textures (e.g., brick joints on building walls) were fully extracted with minimal noise. This indicates that the unpruned model retained more complete low-level features, providing richer semantic information for subsequent detection heads—key to maintaining its high mAP (0.752).

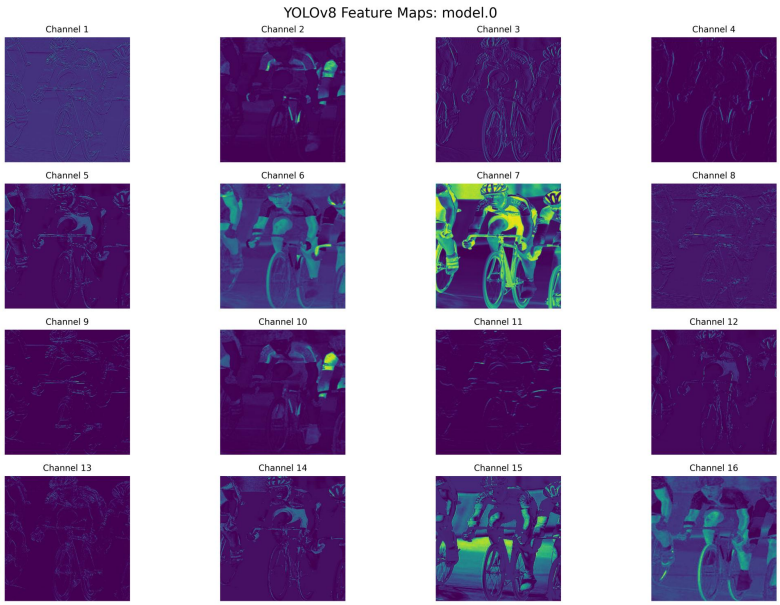

Fig. 6. The feature maps of the pruned model

(3) Feature Recovery of the Distilled Model

In the feature maps of the distilled model at the model.0 layer, channels with weak responses in the pruned model (e.g., Channel 5 and Channel 12) reactivated significantly: Channel 5 recovered clear zebra crossing textures, and Channel 12 improved the continuity of bicycle wheel edges. Additionally, edge response intensities in Channels 3 and 7 approached those of the teacher model (only 5%-8% lower). This suggests that knowledge distillation, guided by the teacher model′s

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"dark knowledge" (e.g., feature distributions and gradient information), help the student model recover critical features lost during pruning, thereby restoring mAP@0.5 from 0.738 to 0.748.

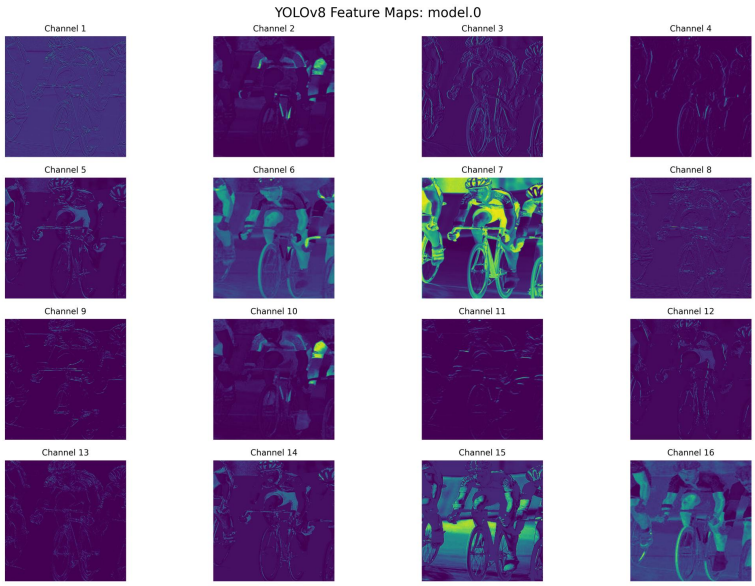

Fig. 7. The feature maps of the distilled model

C. *Validation of Lightweight Strategy Effectiveness*

Combining quantitative metrics and feature visualization results, the "progressive pruning + knowledge distillation" strategy demonstrates the following advantages:

Balance Between Efficiency and Accuracy: The pruned model reduced parameter count by 20.5% and FLOPs (floating-point operations) by 21.7%, while increasing inference speed by 17.6%. The distilled model compensated for feature loss caused by pruning via knowledge distillation, resulting in an mAP@0.5 only 0.5% lower than the original model, achieving a balance between high efficiency and high accuracy.

Interpretability of Feature Preservation: Feature map visualizations directly validated the impact of pruning on feature representation (removal of redundant channels and loss of key features) and the recovery of critical features via distillation (enhanced channel responses and optimized distributions). This provides clear directions for optimizing lightweight strategies — e.g., prioritizing the retention of high-response channels (e.g., Channels 3 and 7 in the model.0 layer) or strengthening constraints on low-level features in distillation loss functions.

D. *Limitations and Improvement Directions*

Several limitations were identified in the experiment, requiring further optimization in future research:

Limitations of the Quantized Model: Although the quantized model achieved the highest inference speed (210 FPS), it did not reduce parameter count (remaining at 11.2M) and suffered more significant accuracy loss for small targets (mAP@0.5:0.95 dropped to 0.475), indicating that INT8 quantization is more sensitive to small-target features.

Robustness of the Pruned-Distilled Model: While the current strategy performs stably in simple scenarios (e.g., single targets, large-scale objects), its robustness in complex scenarios (e.g., dense targets, overlapping small objects) requires further validation. For example, when three pedestrians were densely distributed in the test image, the distilled model′s mAP@0.5 dropped to 0.721 (a 3.6% decrease), suggesting incomplete feature recovery.

Future work may integrate dynamic pruning (adaptively adjusting the pruning rate based on input content) and multi-scale distillation (simultaneously distilling low-level edge features and high-level semantic features) to further enhance the model′s generalization ability and scenario adaptability.

V. Conclusion

This study proposes a lightweight compression method combining "progressive pruning + feature-aware knowledge distillation" to address YOLOv8's high computational complexity and large model size on edge devices. Theoretical analysis, experiments, and feature map visualization validate its effectiveness. Key findings include: First, the progressive pruning strategy, using BN layer scaling factors to evaluate channel importance and an iterative "prune-fine-tune" process, reduces model complexity by retaining critical shallow features, achieving 20.5% parameter reduction (11.2M → 8.9M), 21.7% FLOPs decrease (28.6G → 22.4G), and 17.6% faster inference (142→167 FPS), alleviating edge device burdens. Second, feature-aware distillation compensates

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for pruning-induced accuracy loss: the student model recovers mAP@0.5 from 0.738 (pruned) to 0.748 (only 0.5% lower than the original) via multi-scale feature alignment loss, balancing efficiency and accuracy. Third, feature map visualization reveals pruning-induced feature sparsity (e.g., weakened channel responses) and distillation-enabled recovery (e.g., reactivated key channels), guiding pruning and distillation optimizations. Limitations include the quantized model′s failure to reduce parameters and greater small-target accuracy loss, plus the pruned-distilled model′s unvalidated robustness in complex scenarios (e.g., dense small targets). Future work may integrate dynamic pruning (adaptive pruning rates) and multi-scale distillation (optimizing shallow/deep features) to enhance generalization.

Acknowledgment

This work is supported by the Anhui Provincial Quality Engineering Project of colleges and universities(Grant No.2024xjzlts052),and the Scientific Research Projects Funded by Suzhou University (Grant No.2021bsk016).

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